Forensic Project

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Group 5

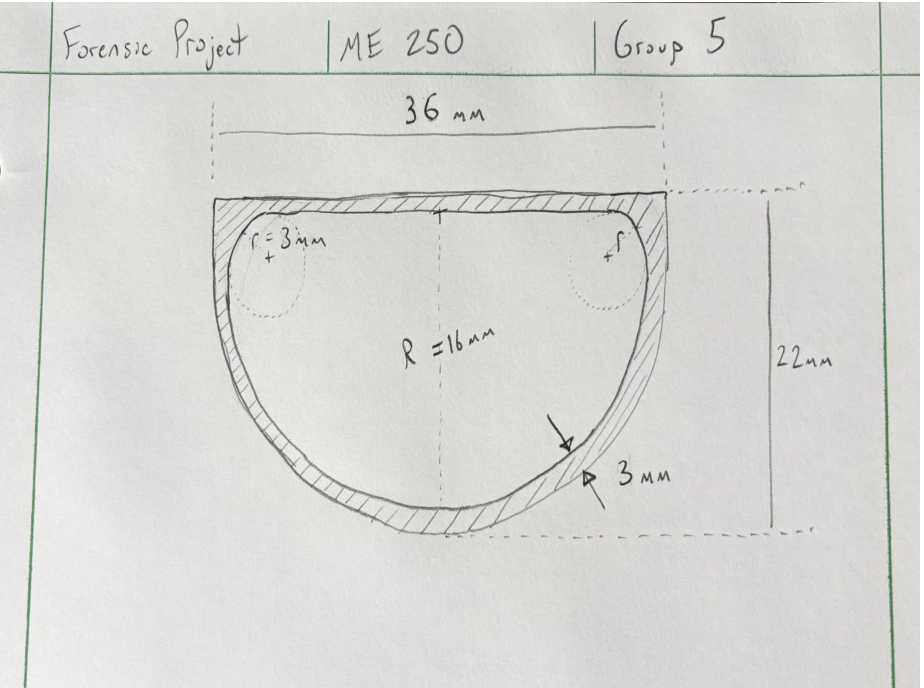
Introduction

Our Forensic Project is focused on this broken refrigerator handle that snapped near where it was connected to the fridge door. This break didn't happen due to a sudden burst of force but do to a small amount of force over time.(Fatigue)

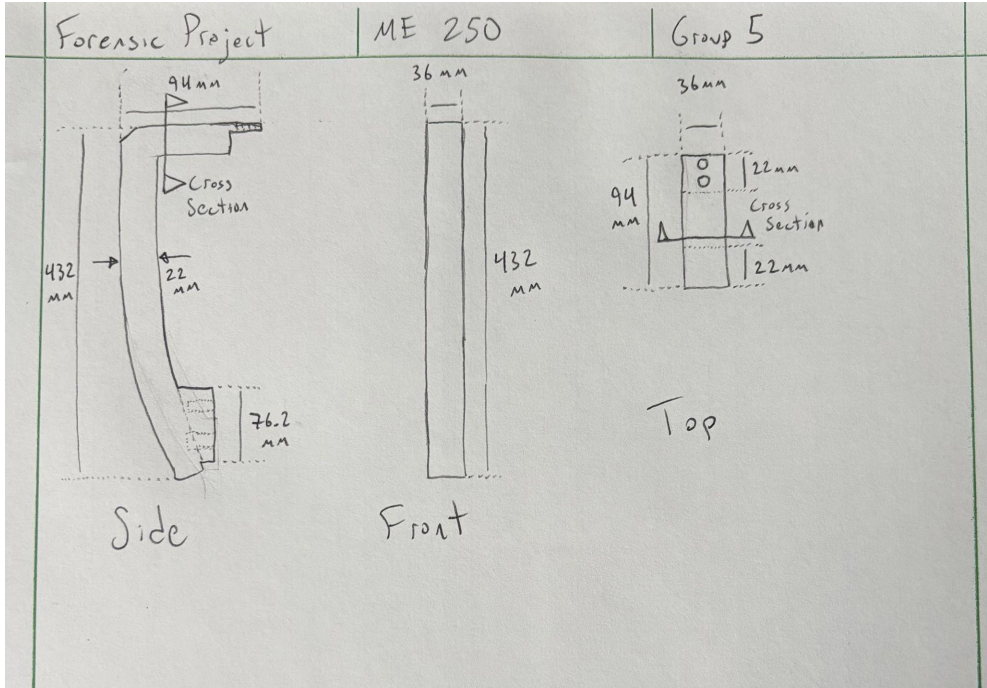
Bolts on the fridge clearly have a higher shear strength than the plastic, causing the weak link to be in the handle at a point near a bend, where the cross sectional area is smallest.

Introduction: Sketches

(Cross Sectional Area Sketch)



(Side View, Front View, and Top View Sketch)



Properties of ABS Plastic

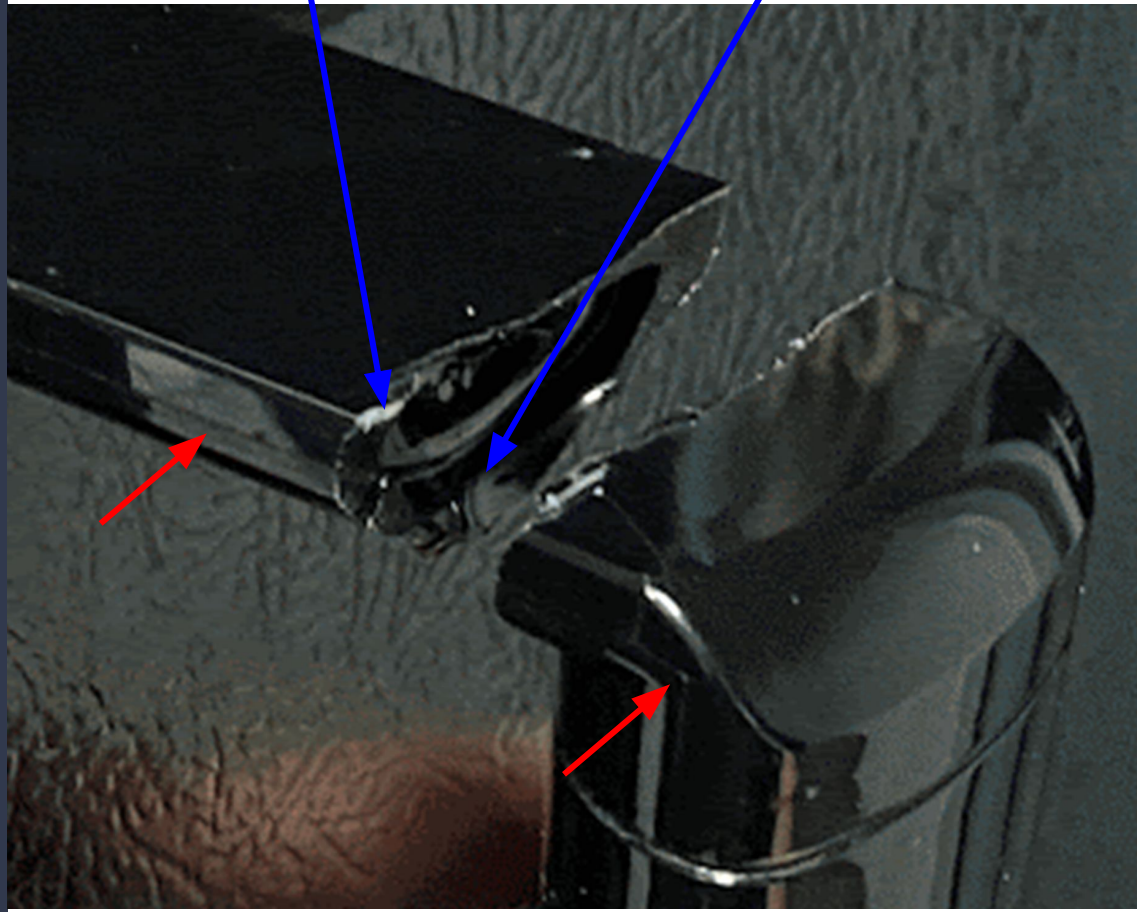
Brittle, not dense, chemical and temperature resistant, recyclable

Due to these properties, it is commonly used in kitchenware which is how we determined that ABS is the material used for the refrigerator handle.

Physical Properties ABS	
Density	0.038 lb/in3
Tensile Strength	5.1 ksi
Tensile Modulus	340 ksi
Elongation at Break Percentage	25%
Flexural Strength	11 ksi =75 MPa
Flexural Modulus	320 ksi
Compressive Strength	9.43 ksi
Compressive Modulus	362.6 ksi
Izod Impact (notched)	7 ft-lb/in
Hardness, Rockwell	R105
Coefficient of Linear Thermal Expansion	5.3 x 10^-5 in./in./°F
Heat Deflection Temp. at 26.4 ksi	215 °F
Water Absorption %, 24hrs	0.3%
Thermal Conductivity	1.2 BTU-in/hr-ft^2-°F

Processing

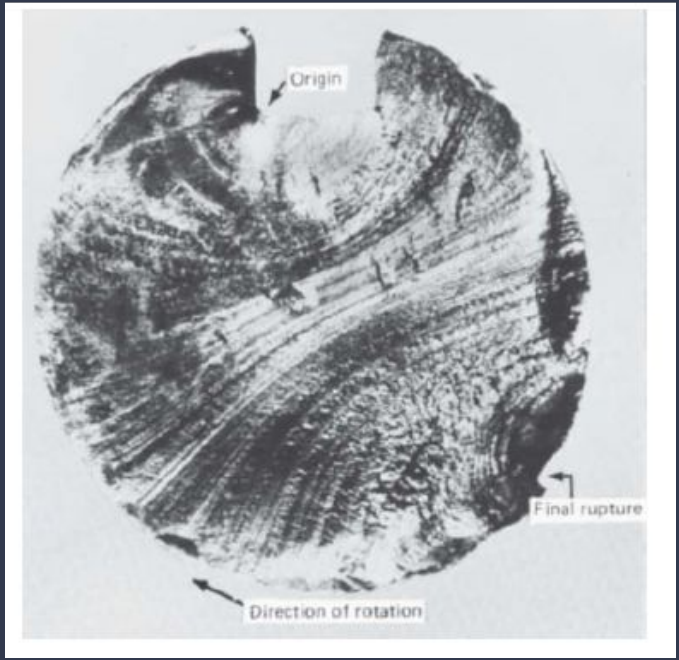
Seems to be Extrusion Blow Molded, closed on bottom end but open on the top with a parison crimp line along the side. Information on this was found by doing research on similar models.



Failure

The handle goes through a daily tensile, compression and torsional stress. The tensile stress is due opening the door and the compression is from closing it shut. We believe torsional stress also has a part to play because you don't pull on the handle and rotate with it; you are most likely pulling out while the door is rotating which would cause a slight torsional stress on the handle.

Excessive force or bending beyond the material's limits can cause damage especially if the plastic is thin or under stress. Therefore, it's important to consider the specific properties and limitations of ABS when attempting to bend or shape it.



From the Textbook

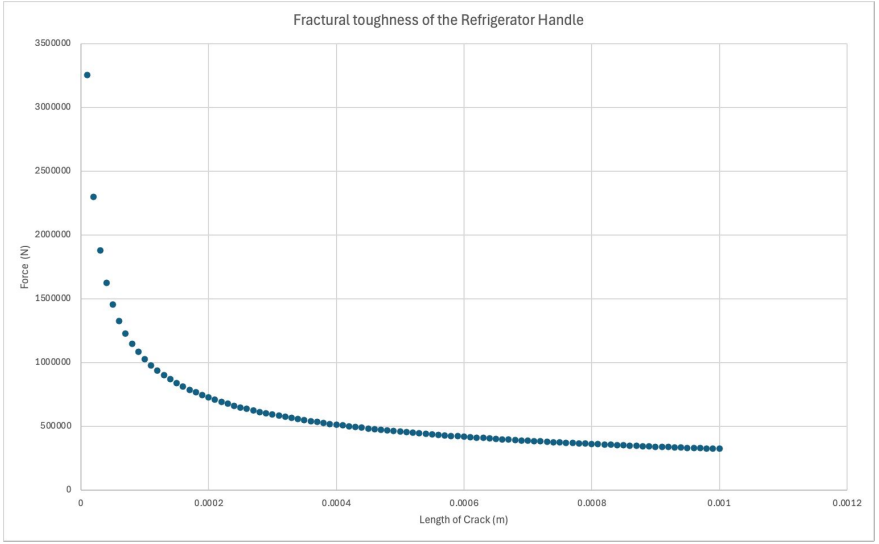
Stresses Involved

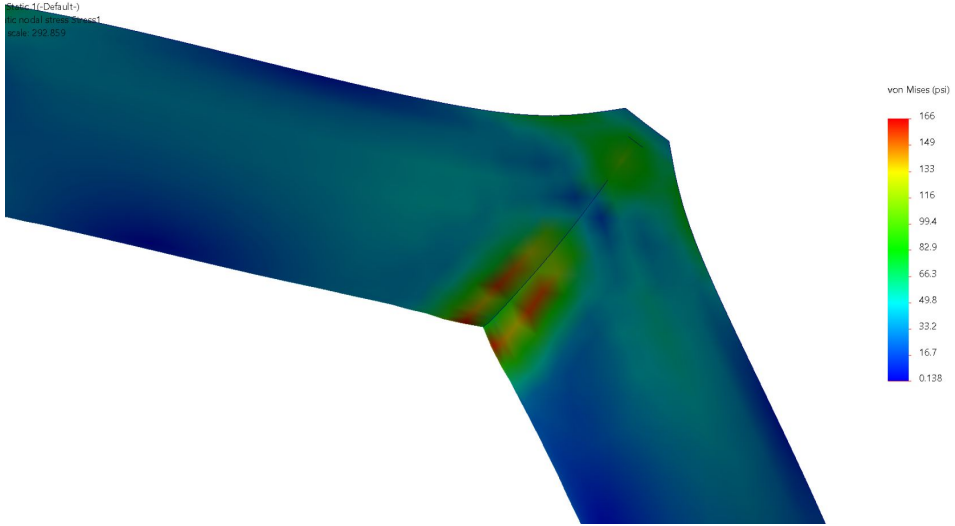
Area	0.000243
Y	1
K	75000000

a value (mm)	a (m)	Force	Stress
0.01	0.00001	3254376	13380930871
0.02	0.00002	2301191	9461746958
0.03	0.00003	1878915	7725484040
0.04	0.00004	1627188	6690465436
0.05	0.00005	1455401	5984134206
0.06	0.00006	1328594	5462742153
0.07	0.00007	1230039	5057516485
0.08	0.00008	1150596	4730873479
0.09	0.00009	1084792	4460310290
0.1	0.0001	1029124	4231421877
0.11	0.00011	981231.3	4034502459
0.12	0.00012	939457.5	3862742020
0.13	0.00013	902601.6	3711202490
0.14	0.00014	869768.6	3576204203
0.15	0.00015	840276.3	3454941495
0.16	0.00016	813594	3345232718
0.17	0.00017	789302.2	3245352432

As we can see from this graph to the right on the fracture toughness on ABS plastic. We can see that as we continue to use the handle, when the crack appears and continues to increase in size, the force needed to break it decreases.

Crack(m) / Tensile Force (N)





Synapsis of the FEA analysis:
FEA Results show that higher stress concentrations happen exactly where the fracture happened.
Applied pressure was 8 lbf (or 35 N)

(FEA ANALYSIS)

Potential Changes in the design

They most likely used ABS plastic because it is very recyclable, chemically resistant and slightly thermally resistant. So we found some other plastics that could be a good replacement that are still recyclable and cost efficient. For example, we could use a more high impact, higher density ABS plastic or a Nylon/fiber plastic which would allow for better performance and added longevity and strength.

Typical Flexural Strength and Flexural Modulus of Polymers

Polymer Type	Flexural Strength (MPa)	Flexural Modulus (GPa)
ABS	75	2.5
ABS + 30% Glass Fiber	120	7
Acetal Copolymer	85	2.5
Acetal Copolymer + 30% Glass Fiber	150	7.5
Acrylic	100	3
Nylon 6	85	2.3
Polyamide-Imide	175	5
Polycarbonate	90	2.3
Polyethylene, MDPE	40	0.7
Polyethylene Terephthalate (PET)	80	1
Polyimide	140	3
Polyimide + Glass Fiber	270	12
Polypropylene	40	1.5
Polystyrene	70	2.5

Nylon 6 %30 glass fiber: about 195 MPa

Flexural Strength Testing of Plastics

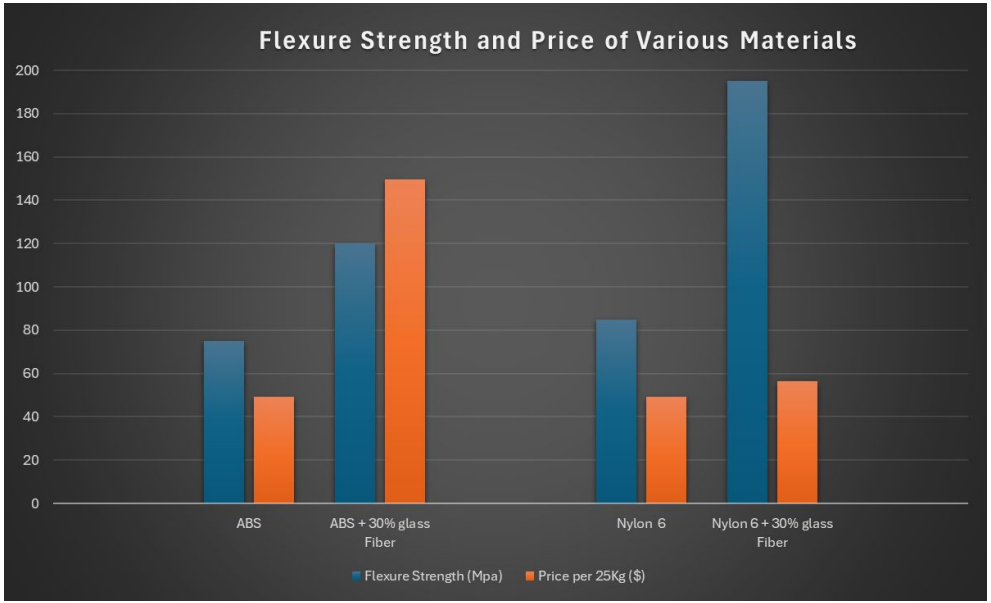
Conclusion

Based on the prices, **Nylon 6 with 30% Glass Fiber** is a great option because it is only 14% more expensive and the Flexural Strength is %160 stronger based off of values for normal ABS. This means you could reduce the amount of material used in each piece to keep the price constant per piece, but you would still increase the Flexure Strength by a noticeable amount.

Prices of plastics per 25 Kg (Not including shipping)

ABS : $49.25 → This Is what was used

ABS + 30% Glass Fiber: $70

Nylon 6: $49.25

Nylon 6 with 30% Glass Fiber: $56.25